

PREFINAL.pptx Review vs PS-26127 + Arcitecture.md

Slide-by-slide gap analysis vs PS-26127 • verdict, fixes & judge prep

07 September 2026 • PPT_REVIEW_PREFINAL.md • City-Wide AI Engine

Stack	Source	Team
PREFINAL.pptx + main.py + api.py	docs/ • Markdown → PDF	SIH 26127 • Prefinal

PS: City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics

Reviewed: 2026-09-07 — Slides 1-8 extracted from docs/PREFINAL.pptx vs problem statement in docs/Arcitecture.md (lines 1-8)

PS demands 3 core functions + 4 solution components:

- 1 High-Accuracy ANPR/OCR Engine >90% in lighting/weather/angle/blur/dirty plates, multi-lane
- 2 Single Plate Trajectory Tracking — spatial-temporal route + timestamps + direction on GIS map
- 3 Macro Traffic Flow Analytics — density, OD patterns, bottlenecks, real-time heatmaps
- 4 Expected: (a) OCR module, (b) Trajectory Reconstruction Engine with query interface, (c) GIS Analytics Dashboard (heatmaps, avg speeds, route densities, flow trends), (d) Alert System (blacklist + route anomaly, real-time)

Verdict

PPT tells a coherent single-camera → city-wide story and maps well to *silos / manual search / no intelligence / bad OCR*. It does **not** yet prove it solves the PS. Approximate coverage:

PS component	PPT coverage	Rating
(a) High-precision OCR >90%, multi-lane, robust	EasyOCR + quality-gate (sharpness/contrast/size) + temporal voting + cost-aware track selection	Partial — approach sound, zero evidence
(b) Trajectory Reconstruction, query-based, chronological GIS path	"single plate query → sightings across network", GIS dashboard shows locations/trajectory	Partial — no cross-camera identity logic
(c) Analytics Dashboard: heatmaps, avg speed, route density, flow trends, OD, bottlenecks	Mentions density, flow-direction, trajectory, congestion	Partial — no speed/direction method, no OD/heatmap implementation
(d) Alert System: blacklist + anomaly, real-time	Not mentioned on any slide	Missing
Vehicle observation moat (Arcitecture.md core): plate + color/make/model + embedding + speed/direction + track ID	Only type (car/bike/bus/truck); no color/make/model, no ReID embedding, no speed/direction	Missing
Enterprise pipeline: FFmpeg → YOLO → ByteTrack → plate → OCR → Redis/NATS → Bun API → Postgres/PostGIS/pgvector → React/MapLibre	Slide 3 is a component table, not a data-flow diagram; no streaming, no DB schema, no calibration/time-sync	Weak

Score: ~55% problem solved on paper, ~30% proven.

What the PPT does well

- Slide 2 "HOW IT ADDRESSES THE PROBLEM" is the strongest slide — directly kills the 4 pain points from the PS (silos → common identity+camera+timestamp; manual search → single query; no intelligence → aggregated analytics; bad OCR → gating+voting).
- Innovations are real and defensible: temporal OCR voting, quality-gated OCR, cost-aware track selection (don't OCR every frame). This matches Arcitecture.md §6 exactly.
- Feasibility is credible: open-source stack, reuse existing CCTV, CPU pilot → GPU scale, incremental rollout prototype → 1-cam → junction → zone → city.
- Challenges → strategies table (Slide 5/6) is honest: low-quality video, OCR errors, compute load, coverage/sync.
- Impact slide (Slide 7) covers operational + traffic + social/economic/environmental — judges like this.
- References are correct: ByteTrack ECCV 2022, YOLO11 docs, EasyOCR, OpenCV.

Where it fails / gaps

P0 — will lose marks if not fixed

- 1 **Alert System completely absent.** PS explicitly requires blacklist + suspicious-route anomaly in real time. Add a block: watchlist DB → match on plate/embedding → push alert + GIS pin + evidence clip.
- 2 **No >90% proof.** "High accuracy" on Slide 6 with no number, no dataset, no test conditions. PS demands >90% across lighting/weather/angle/blur/dirty plates. Need: test set size, accuracy %, precision/recall, failure examples (OD02AB1234 vs OD02AB123B consensus example from Arcitecture.md §18).
- 3 **No cross-camera identity.** ByteTrack = within-camera track_id only. Trajectory across CAM-01 → CAM-07 → CAM-12 needs ReID embedding + cosine similarity + fused identity_score (plate 0.40 + embedding 0.25 + color/model + temporal/spatial). Currently PPT implies ByteTrack alone links cameras — technically false, judges will catch it.
- 4 **No speed/direction method.** PPT claims flow-direction/congestion but never says how speed is computed. Pixels/time = wrong. Need one line: zone-calibration (20m line A→B) / homography → km/h; track vector → road heading (N/NE/E...).
- 5 **Missing vehicle attributes.** Arcitecture.md observation event = camera_id + timestamp + lat/lon + plate+conf + type/color/make/model + speed/direction + track_id + embedding + bbox. PPT only has type. Add color + make/model (even as Phase-2) or explicitly scope them out.

P1 — weakens technical credibility

- 1 **Slide 3 is not an architecture.** It's a table with a blank Technology cell for Backend API, truncated "Vehicle, camera and geospatial da/ta storage", no Redis/NATS/Kafka, no FastAPI/gRPC, no Postgres+PostGIS+pgvector, no FFmpeg/RTSP, no MapLibre. Replace with the pipeline diagram from Arcitecture.md §1/§15.
- 2 **Plate detection is hand-waved.** "selective plate detection" — which model? Separate YOLO plate detector + perspective correction + enhancement before EasyOCR (Arcitecture.md §5-6). Why EasyOCR over PaddleOCR? One line justification needed.
- 3 **No multi-lane, no Indian-plate dataset strategy.** Arcitecture.md stresses fine-tuning on Indian traffic CCTV as the moat. PPT says nothing about data, labeling failure cases, or multi-lane handling.
- 4 **Slides 4+5+6 are the same feasibility content thrice.** Wastes 3 of 8 slides. Merge into one; free space for architecture diagram + demo evidence + alert system.

- 5 **Prototype evidence is assertive, not evidentiary.** "Real-time detection", "high accuracy", "trajectory visualization" with no metrics, no screenshots described, no video/QR link, no latency (FPS, OCR ms). Add one quantitative strip: e.g. Test video X min, Y tracks, Z plates, accuracy %, FPS on CPU/GPU.

P2 — polish

- Typos: ARCHTITECHURE , cameranodes , SIH 2025 on Slide 6 vs 2026 elsewhere, truncated PostGIS row, empty backend-tech cell.
- GIS dashboard tech unnamed (React + MapLibre/Mapbox? PostGIS geospatial queries?).
- No OD patterns, bottleneck detection, or heatmap rendering method named despite PS requiring them.
- No scale numbers: streams per node, Kafka/Redis throughput, retention, privacy (plate masking, access control) — enterprise-grade claim needs at least a mention.

Slide-by-slide fixes

- **S1 Title:** fine. Fix Theme/Category formatting, keep PS ID 26127 prominent.
- **S2 Idea:** keep. Add one line under ANPR: "consensus over N frames, e.g. 0.82/0.71/0.97/0.95 → OD02AB1234". Clarify cross-camera = plate + ReID + time/route, not ByteTrack alone.
- **S3 Technical:** rebuild as data-flow diagram + 2-line stack (Python/PyTorch/Ultralytics/ByteTrack/EasyOCR/OpenCV/ONNX + Redis Streams + Bun/Elysia + Postgres/PostGIS/pgvector + React/MapLibre + FFmpeg). Fix blank/truncated cells.
- **S4/S5/S6 Feasibility:** merge into ONE slide: left = why feasible (open-source, existing CCTV, modular), middle = 4 challenges → 4 strategies with ✓ outcome, right = 5-stage roadmap + prototype metrics strip. Delete duplicates.
- **S6 freed slot → NEW: Trajectory + Analytics + Alerts demo.** Show CAM-01 10:32 → CAM-07 10:39 → CAM-12 10:47 map mock + query box + blacklist alert mock + heatmap thumbnail. This single visual proves (b)+(c)+(d).
- **S7 Impact:** keep, trim to 6 bullets. Good as is.
- **S8 References:** keep ByteTrack/YOLO/EasyOCR/OpenCV. Add PostGIS/pgvector + one Indian ANPR dataset or PaddleOCR comparison cite if used.

Minimum additions to claim "solves PS"

- 1 One architecture diagram (end-to-end event flow with observation JSON fields).
- 2 One accuracy box: dataset, conditions tested, accuracy %, temporal-voting gain (single-frame vs voted).
- 3 One trajectory box: identity fusion formula + GIS chronological path example.
- 4 One alerts box: blacklist/anomaly flow + latency target.
- 5 One analytics box: how density/OD/speed/heatmap are computed (PostGIS aggregation + speed calibration).
- 6 Delete duplicate feasibility slides to make room; fix typos.

Likely judge questions to prep

- How do you link the same vehicle across two cameras when plates are unreadable? → ReID embedding cosine + color/model + time/route fusion, not ByteTrack.
- What is your measured OCR accuracy and on what data? → need number + conditions.
- How is speed computed from a fixed CCTV? → calibrated distance/time, homography later; not pixel velocity.
- How does the alert fire in real time? → watchlist lookup on event stream → WebSocket/push + GIS pin.
- What breaks at 100 cameras? → per-node AI worker + Redis/NATS fan-in + GPU pool + pgvector sharding; CPU pilot only for 1-4 streams.

End of document • Generated from Markdown • SIH 26127